Database Management System LAB
Spring Semester 2024-25
Lab Sheet 1

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Ans – 1)

```
mysql> create table Student(
    -> Rno INT,
    -> Name VARCHAR(100),
    -> DOB DATE,
    -> gender CHAR(1),
    -> Class int,
    -> College varchar(50),
    -> City varchar(20),
    -> Marks int
    -> );
Query OK, 0 rows affected (0.04 sec)
```

Ans – 2)

```
mysql> insert into Student(Rno,Name,DOB,gender,Class,College,City,Marks)
    -> Values
    -> (1,'Goku','1973-01-01','M',100,'Dragon Ball','West City',9999),
    -> (2,'Vegeta','1973-01-02','M',90,'Dragon Ball','East District',80),
    -> (3,'Naruto','1997-01-01','M',4,'Naruto Shippuden','Konoha',50),
    -> (4,'Anya Forger','2022-04-09','F',1,'Spy X Family','Forger Residence',60),
    -> (5,'Yagami Light','2006-10-04','M',10,'Death Note','Tokyo',99);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Ans – 3)

```
mysql> select *from Student;
+------+--------------+------------+--------+-------+-----------------+-----------------+-------+
| Rno  | Name         | DOB        | gender | Class | College         | City            | Marks |
+------+--------------+------------+--------+-------+-----------------+-----------------+-------+
|    1 | Goku         | 1973-01-01 | M      |   100 | Dragon Ball     | West City       |  9999 |
|    2 | Vegeta       | 1973-01-02 | M      |    90 | Dragon Ball     | East District   |    80 |
|    3 | Naruto       | 1997-01-01 | M      |     4 | Naruto Shippuden | Konoha          |    50 |
|    4 | Anya Forger  | 2022-04-09 | F      |     1 | Spy X Family    | Forger Residence |    60 |
|    5 | Yagami Light | 2006-10-04 | M      |    10 | Death Note      | Tokyo           |    99 |
+------+--------------+------------+--------+-------+-----------------+-----------------+-------+
5 rows in set (0.00 sec)
```

Ans – 4)

```
mysql> Describe Student;
+---------+--------------+------+-----+---------+-------+
| Field   | Type         | Null | Key | Default | Extra |
+---------+--------------+------+-----+---------+-------+
| Rno     | int          | YES  |     | NULL    |       |
| Name    | varchar(100) | YES  |     | NULL    |       |
| DOB     | date         | YES  |     | NULL    |       |
| gender  | char(1)      | YES  |     | NULL    |       |
| Class   | int          | YES  |     | NULL    |       |
| College | varchar(50)  | YES  |     | NULL    |       |
| City    | varchar(20)  | YES  |     | NULL    |       |
| Marks   | int          | YES  |     | NULL    |       |
+---------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

Ans – 5) Here note tha, instead of Using College name ‘Bhubaneswar’ – ‘Dragon ‘Ball’ is used.

```
mysql> select Rno,Name,Class from Student where College = 'Dragon Ball';
+------+--------+-------+
| Rno  | Name   | Class |
+------+--------+-------+
|    1 | Goku   |   100 |
|    2 | Vegeta |    90 |
+------+--------+-------+
2 rows in set (0.00 sec)
```

Ans – 6)

```
mysql> select * from Student
    -> Order by Marks ASC;
+------+--------------+------------+--------+-------+------------------+------------------+-------+
| Rno  | Name         | DOB        | gender | Class | College          | City             | Marks |
+------+--------------+------------+--------+-------+------------------+------------------+-------+
|    3 | Naruto       | 1997-01-01 | M      |     4 | Naruto Shippuden | Konoha           |    50 |
|    4 | Anya Forger  | 2022-04-09 | F      |     1 | Spy X Family     | Forger Residence |    60 |
|    2 | Vegeta       | 1973-01-02 | M      |    90 | Dragon Ball      | East District    |    80 |
|    5 | Yagami Light | 2006-10-04 | M      |    10 | Death Note       | Tokyo            |    99 |
|    1 | Goku         | 1973-01-01 | M      |   100 | Dragon Ball      | West City        |  9999 |
+------+--------------+------------+--------+-------+------------------+------------------+-------+
5 rows in set (0.00 sec)
```

Ans – 7)

```
mysql> update Student
    -> Set Marks = 89
    -> Where Rno = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> Select * from Student;
+------+--------------+------------+--------+-------+------------------+------------------+-------+
| Rno  | Name         | DOB        | gender | Class | College          | City             | Marks |
+------+--------------+------------+--------+-------+------------------+------------------+-------+
|    1 | Goku         | 1973-01-01 | M      |   100 | Dragon Ball      | West City        |  9999 |
|    2 | Vegeta       | 1973-01-02 | M      |    90 | Dragon Ball      | East District    |    80 |
|    3 | Naruto       | 1997-01-01 | M      |     4 | Naruto Shippuden | Konoha           |    50 |
|    4 | Anya Forger  | 2022-04-09 | F      |     1 | Spy X Family     | Forger Residence |    60 |
|    5 | Yagami Light | 2006-10-04 | M      |    10 | Death Note       | Tokyo            |    89 |
+------+--------------+------------+--------+-------+------------------+------------------+-------+
5 rows in set (0.00 sec)
```

Ans – 8) Rno = 3 is used, because, Rno = 9 is not present;

```
mysql> update Student
    -> Set Name = 'Uzumaki Naruto', City = 'Leaf Village'
    -> Where Rno = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Student;
+------+----------------+------------+--------+-------+------------------+------------------+-------+
| Rno  | Name           | DOB        | gender | Class | College          | City             | Marks |
+------+----------------+------------+--------+-------+------------------+------------------+-------+
|    1 | Goku           | 1973-01-01 | M      |   100 | Dragon Ball      | West City        |  9999 |
|    2 | Vegeta         | 1973-01-02 | M      |    90 | Dragon Ball      | East District    |    80 |
|    3 | Uzumaki Naruto | 1997-01-01 | M      |     4 | Naruto Shippuden | Leaf Village     |    50 |
|    4 | Anya Forger    | 2022-04-09 | F      |     1 | Spy X Family     | Forger Residence |    60 |
|    5 | Yagami Light   | 2006-10-04 | M      |    10 | Death Note       | Tokyo            |    89 |
+------+----------------+------------+--------+-------+------------------+------------------+-------+
5 rows in set (0.00 sec)
```

Ans – 9)

```
mysql> delete from Student
    -> Where City = 'East District';
Query OK, 1 row affected (0.07 sec)

mysql> select * from Student;
+------+---------------+------------+--------+-------+-----------------+-----------------+-------+
| Rno  | Name          | DOB        | gender | Class | College         | City            | Marks |
+------+---------------+------------+--------+-------+-----------------+-----------------+-------+
|    1 | Goku          | 1973-01-01 | M      |   100 | Dragon Ball     | West City       |  9999 |
|    3 | Uzumaki Naruto | 1997-01-01 | M      |     4 | Naruto Shippuden | Leaf Village    |    50 |
|    4 | Anya Forger   | 2022-04-09 | F      |     1 | Spy X Family    | Forger Residence |    60 |
|    5 | Yagami Light  | 2006-10-04 | M      |    10 | Death Note      | Tokyo           |    89 |
+------+---------------+------------+--------+-------+-----------------+-----------------+-------+
4 rows in set (0.00 sec)
```

Ans – 10)

```
mysql> delete from Student
    -> Where Marks < 70;
Query OK, 2 rows affected (0.00 sec)

mysql> select * from Student;
+------+--------------+------------+--------+-------+-------------+-----------+-------+
| Rno  | Name         | DOB        | gender | Class | College     | City      | Marks |
+------+--------------+------------+--------+-------+-------------+-----------+-------+
|    1 | Goku         | 1973-01-01 | M      |   100 | Dragon Ball | West City |  9999 |
|    5 | Yagami Light | 2006-10-04 | M      |    10 | Death Note  | Tokyo     |    89 |
+------+--------------+------------+--------+-------+-------------+-----------+-------+
2 rows in set (0.01 sec)
```

Ans – 11)

```
mysql> insert into Employee(EmpNo,Ename,Job,Salary,DeptNo)
    -> Values
    -> (1,'Goku C','GOD',100000,40),
    -> (2,'Obito Uchiha','Clerk',2500,10),
    -> (3,'Armin Arlert','Useless',4000,50),
    -> (4,'C.C.','Salesperson',1000,20),
    -> (5,'All Might','meh',500,30),
    -> (6,'Misa','Assasin',2750,10);
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> select * from Employee;
+-------+--------------+-------------+--------+--------+
| EmpNo | Ename        | Job         | Salary | DeptNo |
+-------+--------------+-------------+--------+--------+
|     1 | Goku C       | GOD         | 100000 |     40 |
|     2 | Obito Uchiha | Clerk       |   2500 |     10 |
|     3 | Armin Arlert | Useless     |   4000 |     50 |
|     4 | C.C.         | Salesperson |   1000 |     20 |
|     5 | All Might    | meh         |    500 |     30 |
|     6 | Misa         | Assasin     |   2750 |     10 |
+-------+--------------+-------------+--------+--------+
6 rows in set (0.01 sec)
```

Ans – 12)

Ans – 13)

```
mysql> select * from Employee;
+-------+--------------+-------------+--------+--------+
| EmpNo | Ename        | Job         | Salary | DeptNo |
+-------+--------------+-------------+--------+--------+
|     1 | Goku C       | GOD         | 100000 |     40 |
|     2 | Obito Uchiha | Clerk       |   2500 |     10 |
|     3 | Armin Arlert | Useless     |   4000 |     50 |
|     4 | C.C.         | Salesperson |   1000 |     20 |
|     5 | All Might    | meh         |    500 |     30 |
|     6 | Misa         | Assasin     |   2750 |     10 |
+-------+--------------+-------------+--------+--------+
6 rows in set (0.01 sec)

mysql> select EmpNo,EName from Employee where DeptNo=10;
+-------+--------------+
| EmpNo | EName        |
+-------+--------------+
|     2 | Obito Uchiha |
|     6 | Misa         |
+-------+--------------+
2 rows in set (0.00 sec)
```

Ans – 14)

```
mysql> select Ename from Employee where Salary > 2000;
+--------------+
| Ename        |
+--------------+
| Goku C       |
| Obito Uchiha |
| Armin Arlert |
| Misa         |
+--------------+
4 rows in set (0.00 sec)
```

Ans – 15)

```
mysql> select Ename,Salary from Employee where Job='Salesperson' or Job='Clerk';
+--------------+--------+
| Ename        | Salary |
+--------------+--------+
| Obito Uchiha |   2500 |
| C.C.         |   1000 |
+--------------+--------+
2 rows in set (0.00 sec)
```

Ans – 16)

```
mysql> select * from Employee where Salary Between 2000 and 3000;
+-------+--------------+---------+--------+--------+
| EmpNo | Ename        | Job     | Salary | DeptNo |
+-------+--------------+---------+--------+--------+
|     2 | Obito Uchiha | Clerk   |   2500 |     10 |
|     6 | Misa         | Assasin |   2750 |     10 |
+-------+--------------+---------+--------+--------+
2 rows in set (0.00 sec)
```

Ans – 17)

```
mysql> select * from Employee where DeptNo=10 or DeptNo=20 or DeptNo=30;
+-------+--------------+-------------+--------+--------+
| EmpNo | Ename        | Job         | Salary | DeptNo |
+-------+--------------+-------------+--------+--------+
|     2 | Obito Uchiha | Clerk       |   2500 |     10 |
|     4 | C.C.         | Salesperson |   1000 |     20 |
|     5 | All Might    | meh         |    500 |     30 |
|     6 | Misa         | Assasin     |   2750 |     10 |
+-------+--------------+-------------+--------+--------+
4 rows in set (0.00 sec)
```

Ans – 18)

```
mysql> alter table Employee add column Commision int default 1000;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> update Employee set Commision = NULL where Salary > 2600;
Query OK, 3 rows affected (0.07 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> select Ename from Employee where Commision IS NULL;
+--------------+
| Ename        |
+--------------+
| Goku C       |
| Armin Arlert |
| Misa         |
+--------------+
3 rows in set (0.00 sec)
```

Ans – 19)

```
mysql> select DeptNo,Salary from Employee
    -> order by DeptNo ASC, Salary DESC;
+--------+--------+
| DeptNo | Salary |
+--------+--------+
|     10 |   2750 |
|     10 |   2500 |
|     20 |   1000 |
|     30 |    500 |
|     40 | 100000 |
|     50 |   4000 |
+--------+--------+
6 rows in set (0.00 sec)
```

Ans – 20)

```
mysql> select Ename from Employee where Ename like 'C%';
+-------+
| Ename |
+-------+
| C.C.  |
+-------+
1 row in set (0.00 sec)
```

Ans – 21)

```
mysql> select Ename from Employee where Ename like '%C';
+--------+
| Ename  |
+--------+
| Goku C |
+--------+
1 row in set (0.00 sec)
```

Ans – 22)

```
mysql> select Ename from Employee where LOWER(Ename) LIKE '%a%a%';
+--------------+
| Ename        |
+--------------+
| Armin Arlert |
+--------------+
1 row in set (0.00 sec)
```

Ans – 23)

```
mysql> select Ename from Employee where LOWER(Ename) LIKE '_b%';
+--------------+
| Ename        |
+--------------+
| Obito Uchiha |
+--------------+
1 row in set (0.00 sec)
```

Ans – 24)

```
mysql> select Ename from Employee where LOWER(Ename) LIKE 'a%' or LOWER(Ename) LIKE '%a';
+--------------+
| Ename        |
+--------------+
| Obito Uchiha |
| Armin Arlert |
| All Might    |
| misA         |
+--------------+
4 rows in set (0.00 sec)
```